ReviewGrounder: Improving Review Substantiveness with Rubric-Guided, Tool-Integrated Agents

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ACL 2026

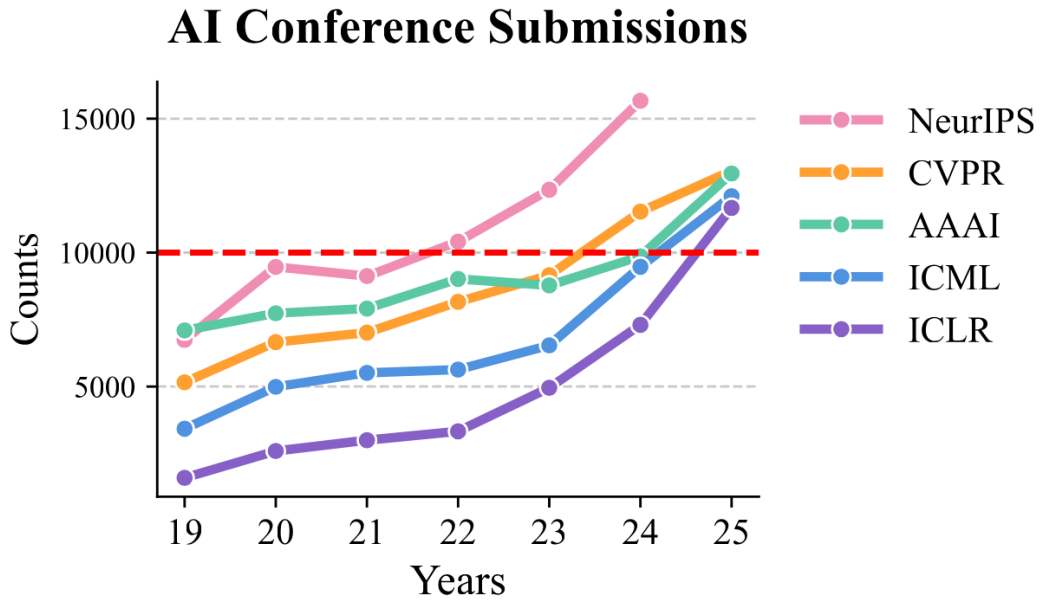
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Background

- Paper submissions have increased in recent years
- LLMs could help support the peer review workflow



Position: The AI Conference Peer Review Crisis Demands Author Feedback and Reviewer Rewards (ICML 2025)

Prior Work

- LLM-based paper reviewers
  - Multi-agent frameworks
  - Supervised fine-tuning (SFT)
  - Reinforcement learning (RL)

⇒ Superficial and formulaic

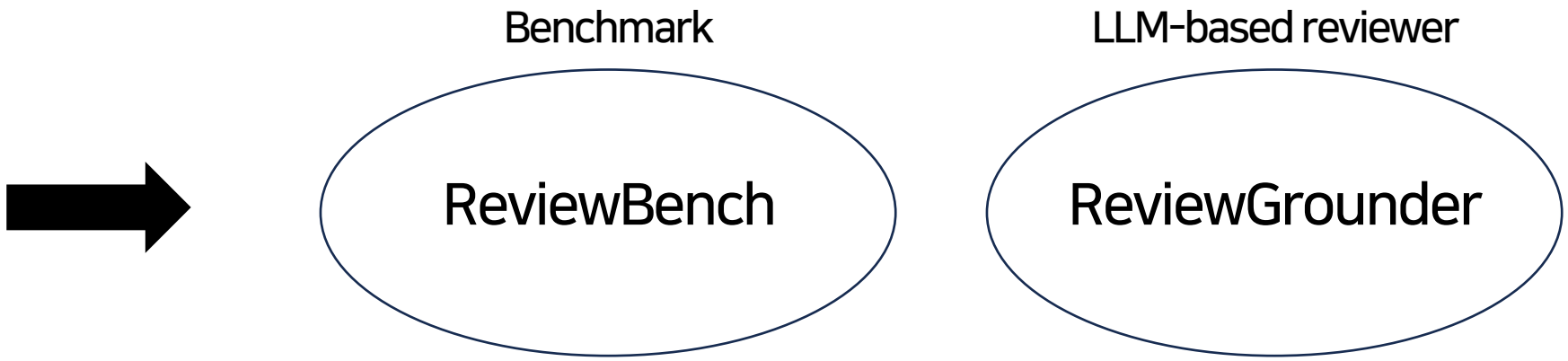
- Automated review evaluation methods
  - Similarity-based evaluation (e.g., ROUGE)
  - LLM-as-a-judge (e.g., direct scoring)

⇒ Coarse-grained and ambiguous

Source of Shortcomings

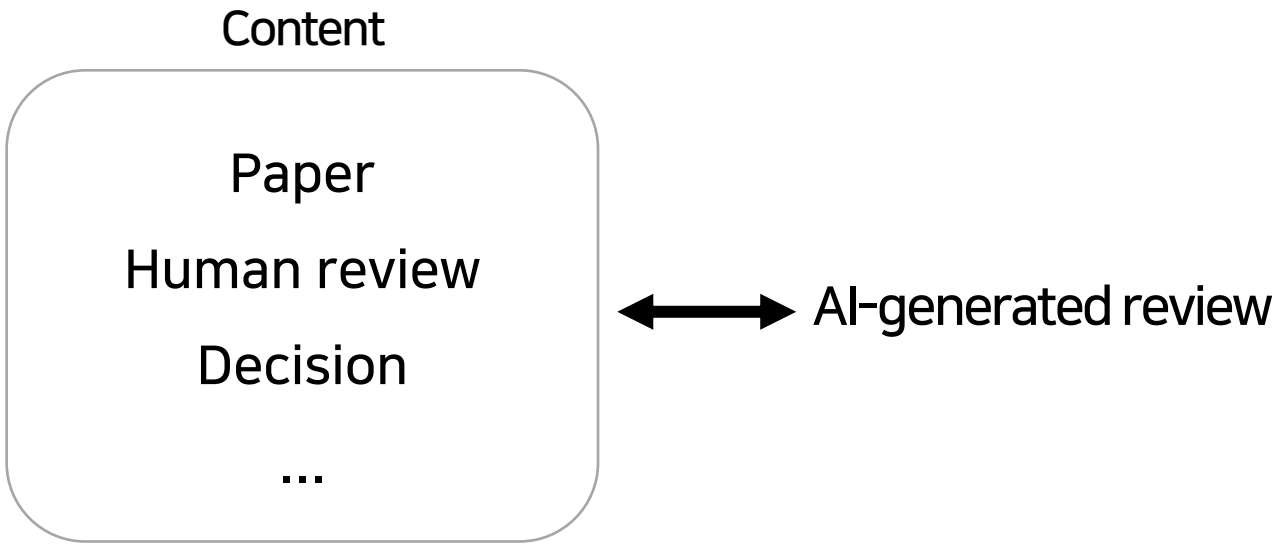
Underutilization of two sources

1. Reviewer guidelines and rubrics
2. Context from existing work



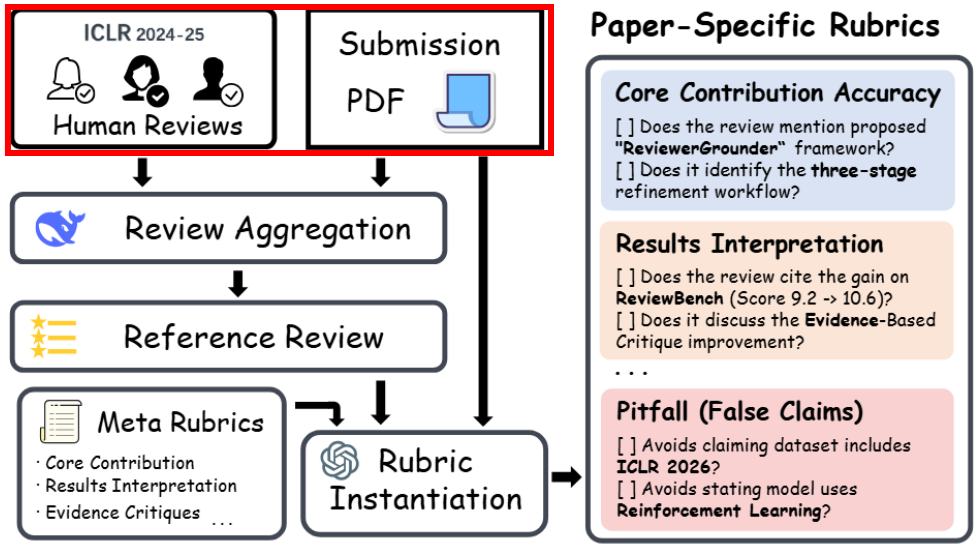
ReviewBench

- Built to evaluate fine-grained evidence grounding in AI-generated peer reviews



Dataset Construction

- Built on DeepReview-13K (ICLR 2024-2025)
- Data filtering

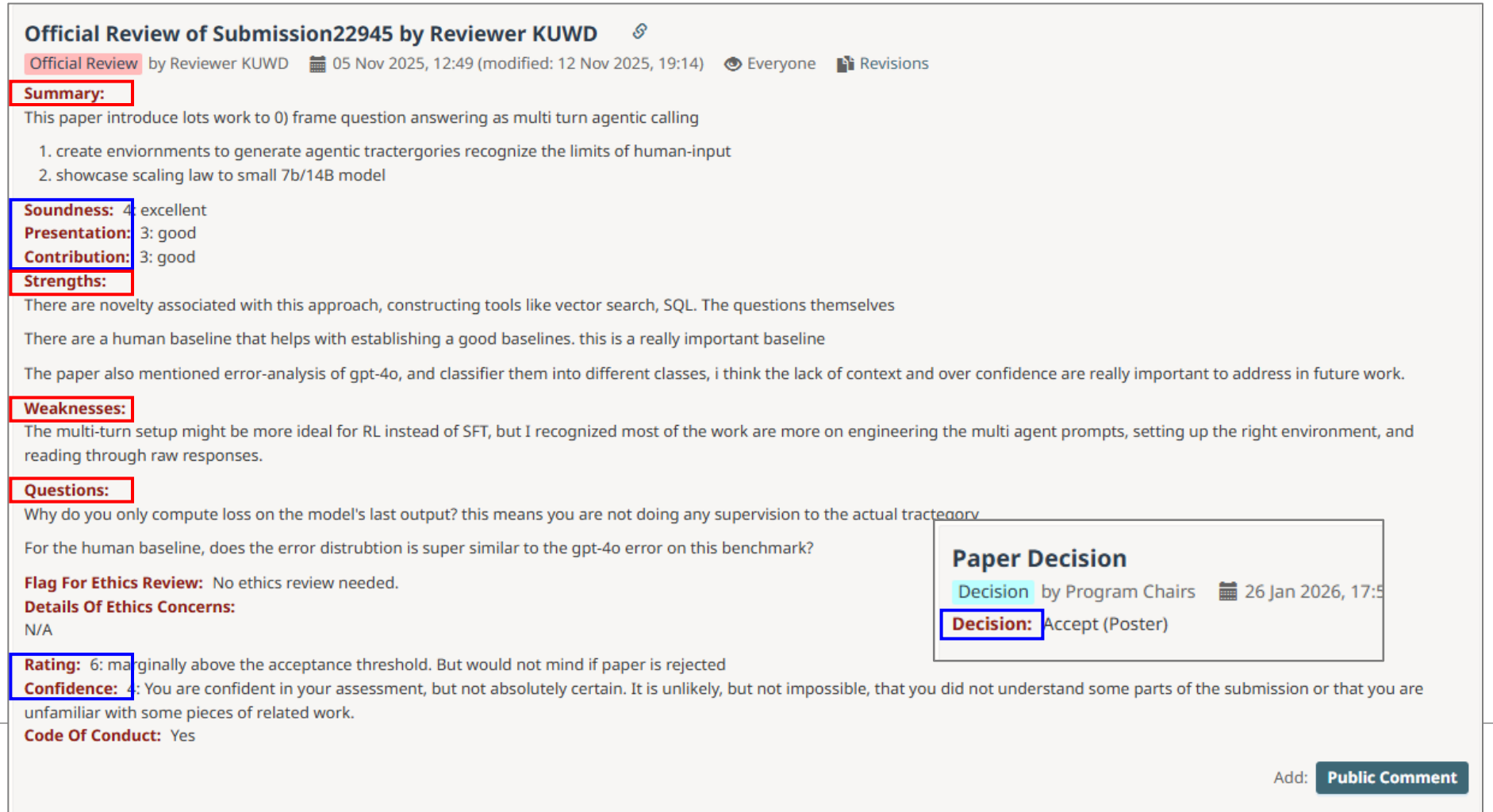


Dataset Construction

- Human review normalization

Textual assessments

Scores and decisions



Official Review of Submission22945 by Reviewer KUWD

Official Review by Reviewer KUWD 05 Nov 2025, 12:49 (modified: 12 Nov 2025, 19:14) Everyone Revisions

Summary:

This paper introduce lots work to 0) frame question answering as multi turn agentic calling

1. create enviornments to generate agentic tractergories recognize the limits of human-input
2. showcase scaling law to small 7b/14B model

Soundness: 4: excellent
Presentation: 3: good
Contribution: 3: good

Strengths:

There are novelty associated with this approach, constructing tools like vector search, SQL. The questions themselves

There are a human baseline that helps with establishing a good baselines. this is a really important baseline

The paper also mentioned error-analysis of gpt-4o, and classifier them into different classes, i think the lack of context and over confidence are really important to address in future work.

Weaknesses:

The multi-turn setup might be more ideal for RL instead of SFT, but I recognized most of the work are more on engineering the multi agent prompts, setting up the right environment, and reading through raw responses.

Questions:

Why do you only compute loss on the model's last output? this means you are not doing any supervision to the actual tractegory

For the human baseline, does the error distrubtion is super similar to the gpt-4o error on this benchmark?

Flag For Ethics Review: No ethics review needed.
Details Of Ethics Concerns:
N/A

Rating: 6: marginally above the acceptance threshold. But would not mind if paper is rejected
Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.
Code Of Conduct: Yes

Paper Decision

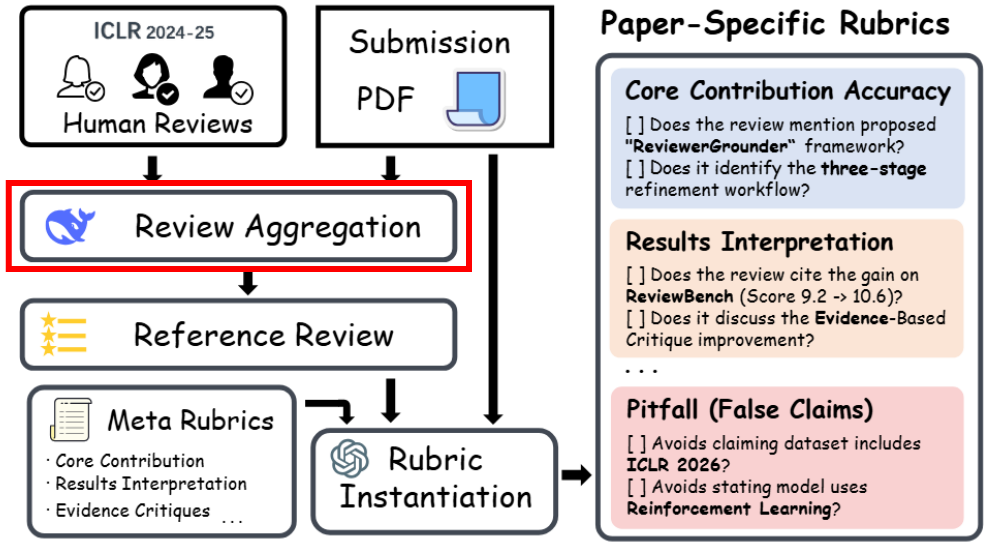
Decision by Program Chairs 26 Jan 2026, 17:5

Decision: Accept (Poster)

Add: Public Comment

Dataset Construction

- Reference review aggregation
  - Ground-truth rating: mean of the human overall ratings
  - Ground-truth decision



Evaluation Approach

1. Rubric-based evaluation
2. Numeric-field evaluation

1. Rubric-based Evaluation

- Paper-agnostic meta-rubrics
- Paper-specific rubrics
- Evaluator

Meta Rubrics

Purpose: These rubrics define the evaluation criteria used to assess the *quality, correctness, and usefulness* of a peer review. Each dimension focuses on a specific aspect of review quality and guides both automated refinement agents and human supervisors.

Evaluation Dimensions:

- **Core Contribution Accuracy**
  Assesses whether the review accurately captures the paper's main contributions and central methodological innovations, especially in the *Summary* and *Strengths*, without misinterpretation.

⋮



Paper-Specific Review Rubrics:
Paper *SqueezeLLM*

Purpose: These rubrics are tailored to the paper *SqueezeLLM: Dense and Sparse Quantization*. It specifies how a review should be evaluated *with respect to this paper's concrete claims, methods, and results*. This rubric is for **display and guidance only**, not for automated scoring.

Paper-Aware Evaluation Dimensions:

- **Core Contribution Accuracy**. Checks whether the review correctly captures the paper's two primary technical contributions: (i) sensitivity-based non-uniform quantization using Fisher-information-weighted k-means, and (ii) dense-and-sparse decomposition for handling outliers and sensitive weights. The review should also accurately reflect the paper's central claim that *memory bandwidth, not compute*, is the main bottleneck for single-batch LLM inference.

⋮

1. Rubric-based Evaluation

- Paper-agnostic meta-rubrics
- Paper-specific rubrics
- Evaluator
  - Eval(Paper, Candidate review, Paper-specific rubrics)
  - Overall score: sum of the individual dimension scores



Scoring Rules

Positive dimension $(0/1/2)$**.** Let $\hat{R}$ denote the paper-specific rule and let $\mathcal{K}(\hat{R})$ be its checklist of key points.

- **Score 0 (not satisfied):** The review addresses none of the key points, *or* it makes a *material* incorrect claim relevant to this dimension.
- **Score 1 (partially satisfied):** The review correctly addresses at least half of the key points and contains no material errors.
- **Score 2 (fully satisfied):** The review correctly addresses all key points (or all but a minor omission that does not affect the criterion) and contains no material errors.

Negative pitfall dimension $(-2/-1/0)$**.**

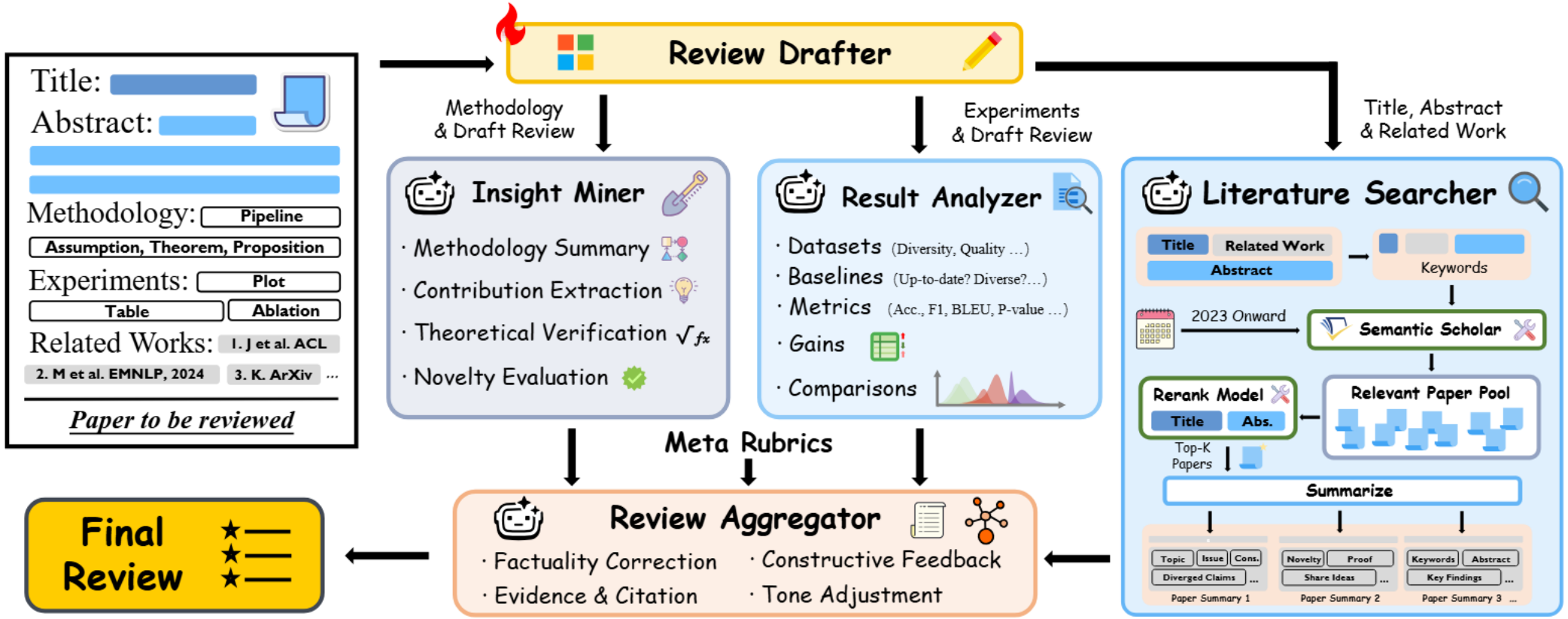
- **Score 0 (none):** The review exhibits none of the pitfall key points.
- **Score** -1 **(mild):** The review exhibits at least one pitfall key point.
- **Score** -2 **(severe):** The review exhibits multiple pitfall key points, *or* a single severe instance (e.g., a clear hallucination of nonexistent content or a direct contradiction of stated results/design choices).

2. Numeric-field Evaluation

- Evaluate the numerical rating and final decision in a candidate review
  - Mean squared error (MSE) and mean absolute error (MAE)
  - Accuracy and F1 score

ReviewGrounder

- Framework for producing grounded and substantive reviews



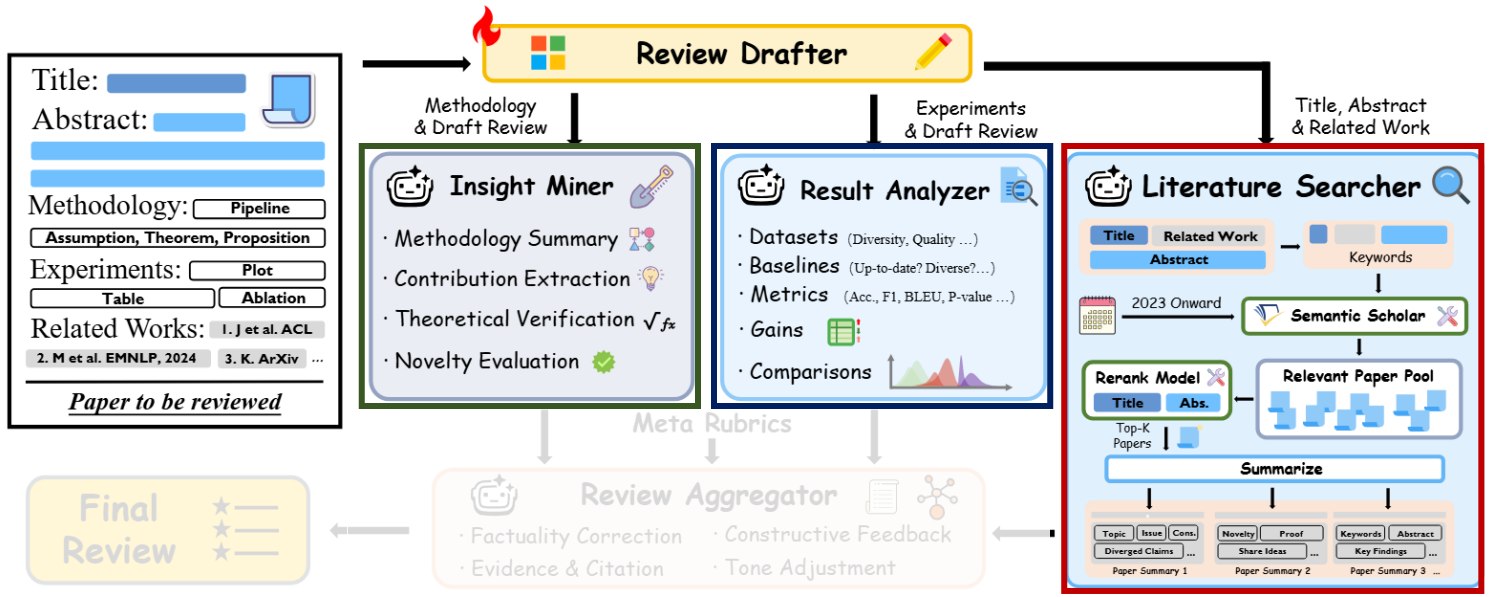
Stage 1: Draft Review Generation

- Produce a draft review using fine-tuned LLM



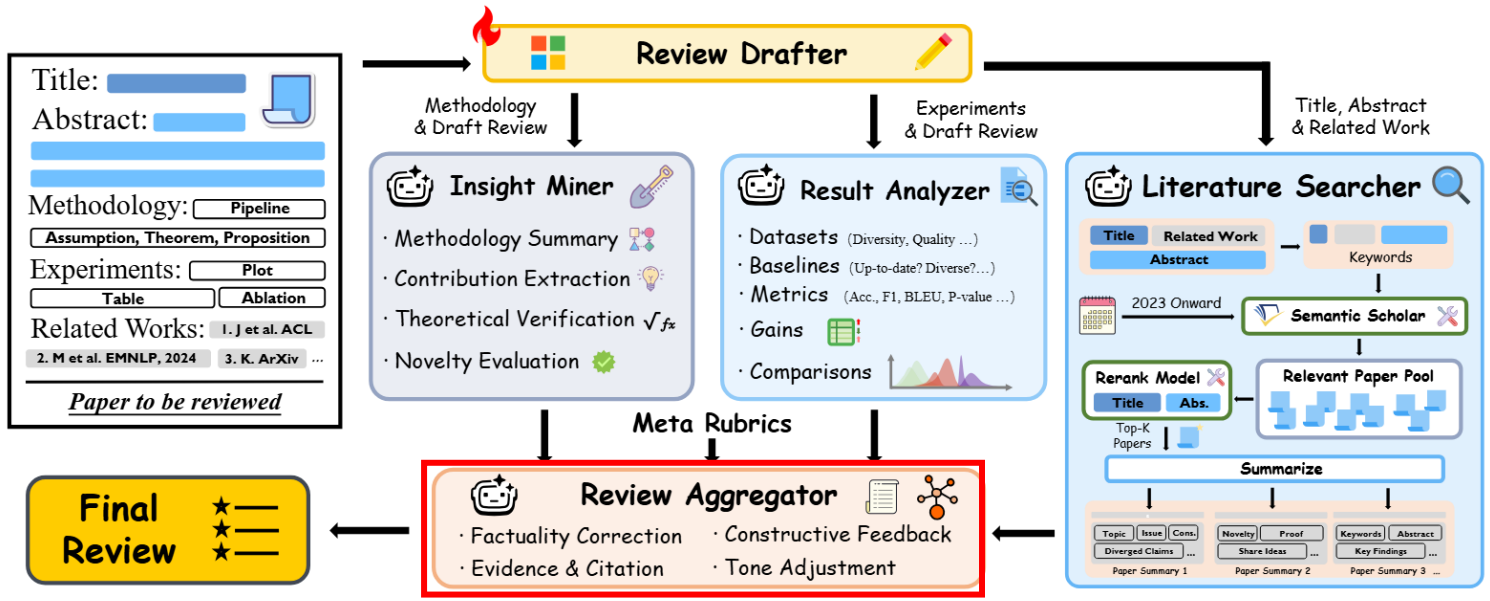
Stage 2: Multi-dimensional Review Grounding

- Enrich and consolidate the draft using three agents
  - Literature searcher: assesses novelty
  - Insight miner: analyzes the paper's core contributions and technical claims
  - Result analyzer: examines experimental design, results, and quantitative evidence



Stage 3: Rubric-guided Review Synthesis

- Synthesizes the outputs of all upstream agents to produce final review
  - Factuality correction
  - Constructive feedback
  - ...



Review from ReviewGrounder

Review from REVIEWGROUNDER

Summary:
The paper proposes SEA (Sparse linear Attention with Estimated Attention mask), a test-time linear-complexity attention module. SEA first estimates a compressed $T \times K$ attention matrix using Performer-based kernel attention and a 3-layer CNN decoder, then generates a sparse mask

⋮

Strengths:
1. Introduces a two-phase linear-attention pipeline (kernel-based estimation + sparse mask) with provable $\mathcal{O}(T)$ inference cost (Section 3.1, Fig. 1).

2. Provides a concrete knowledge-distillation training scheme (Eq. 1) that preserves the

⋮

Weaknesses:
1. **Adaptation cost not fully quantified** – While Section 3.3 describes replacing the teacher's

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Questions:
1. What is the exact training budget (GPU hours, dataset size) required to replace a pretrained

⋮

Rating: 5.5
Decision: reject

Experiments

- Evaluation
  - Rubric-based evaluation: assesses the textual quality
  - Numeric-field evaluation: measures predicted ratings and decisions
- Baselines
  - Foundation models
  - Agentic reviewing frameworks
  - Fine-tuned reviewer models

Main Results

- Overall ReviewGrounder outperforms all baselines

Method	Model	Core	Res.	Comp.	EBC	Clr.	Cov.	Tone	Contradict.	Overall	Δ
Foundation Model	Qwen3-32B	1.6971	0.7642	0.5800	0.1437	1.6128	1.1537	1.9992	-0.1460	7.8047	↑ 38%
	QWQ-32B	1.6901	0.6531	0.3513	0.1186	1.6792	0.9461	1.9969	-0.0836	7.3517	↑ 46%
	GPT-4o	1.1969	0.1037	0.0302	0.0024	1.0499	0.3318	1.9840	-0.1233	4.5756	↑ 135%
	GPT-4.1	1.7573	0.6966	0.3406	0.1074	1.6327	1.1675	1.9992	-0.0397	7.6616	↑ 41%
AgentReview	GPT-4o	1.1300	0.1600	0.1100	0.1250	1.3400	0.5900	2.0000	-0.1600	4.8675	↑ 121%
	GPT-4.1	1.0300	0.1300	0.1200	0.0000	1.4100	0.6300	1.9800	-0.1600	4.9620	↑ 117%
AI Scientist	GPT-4o	0.8500	0.0000	0.0200	0.0000	0.6700	0.1800	1.7600	-0.1900	3.6800	↑ 193%
	GPT-4.1	1.6700	0.4800	0.3600	0.0830	1.5600	1.1300	1.9400	-0.0900	7.0893	↑ 52%
CycleReviewer	Llama-3.1-8B	0.9852	0.1011	0.0645	0.0089	0.5832	0.1493	1.6571	-0.4504	3.0989	↑ 248%
	Llama-3.1-70B	1.0187	0.1633	0.0980	0.0109	0.7698	0.2551	1.8476	-0.6412	3.5220	↑ 206%
DeepReviewer	Phi-4-7B	1.4207	0.4545	0.3299	0.1311	1.3743	1.0599	1.9432	-0.3953	6.3183	↑ 70%
	Phi-4-14B	1.6306	0.6532	0.4977	0.3532	1.6772	1.2877	1.9930	-0.1922	7.9004	↑ 36%
REVIEWGROUNDER	**Phi-4-14B**	**1.8507**	**1.4075**	**0.9059**	**1.4831**	**1.9191**	**1.3289**	**1.9992**	**-0.1245**	**10.7699**	-

Main Results

- ReviewGrounder achieves the lowest rating error and the highest decision prediction accuracy, compared with all baselines

Method	Model	Decision		Rating	
		ACC↑	F1↑	MSE↓	MAE↓
AgentReview	Claude-3-5-sonnet	0.2826	0.2541	2.8406	1.2989
	Gemini-2.0-Flash-Thinking	0.4242	0.4242	2.6186	1.2170
	DeepSeek-V3	0.3140	0.2506	1.9951	1.1017
AI Scientist	GPT-o1	0.4167	0.4157	4.3072	1.7917
	Claude-3-5-sonnet	0.5579	0.4440	3.0992	1.3500
	Gemini-2.0-Flash-Thinking	0.6139	0.4808	3.9232	1.6470
	DeepSeek-V3	0.4059	0.3988	4.8006	1.8403
	DeepSeek-R1	0.4259	0.4161	4.7719	1.8099
CycleReviewer	Llama-3.1-8B	0.2354	0.3988	3.1324	1.3663
	Llama-3.1-70B	0.1545	0.4156	1.8440	1.0643
DeepReviewer	Phi-4-7B	0.6381	0.6068	1.4442	0.9416
	Phi-4-14B	0.6667	0.5204	1.3527	0.9041
REVIEWGROUNDER	**Phi-4-14B**	**0.6939**	**0.6699**	**1.1607**	**0.8597**

Ablation Study

- Proposed framework benefit smaller models
- Omitting any agent leads to a performance degradation

Drafter Phi-4-14B	Drafter Phi-4-7B	Drafter Qwen3-4B	Searcher	Miner	Analyzer	Overall
		✓	✓	✓	✓	10.6418
	✓		✓	✓	✓	10.5928
✓				✓	✓	10.6568
✓			✓		✓	10.6526
✓			✓	✓		10.0186
✓			✓	✓	✓	**10.7699**

Conclusion

- Introduce **ReviewBench**, enabling multi-faceted, rubric-based evaluation
- Present **ReviewGrounder**, a rubric-guided, tool-integrated multi-agent framework for producing peer reviews
- Extensive experiments demonstrate effectiveness of ReviewGrounder

Opinion

- 장점
  - 벤치마크 측면: 텍스트의 정성적인 부분을 어떻게 평가할 것인가를 잘 정의함
  - 리뷰 생성 프레임워크 측면: 논문의 각 섹션별로 에이전트를 두어 병렬적으로 분석
- 아쉬운점
  - 논문의 figure 처리 방법에 대한 구체적인 설명이 없음

Open Question

- What errors might models make when assessing evidence in figures and tables?